

Kaunas University of Technology
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Fault Detection, Analysis, and Rectification Report

Control Systems of Technological Equipment, their Reliability and Diagnostics
FESTO Processing Station Diagnostics

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1. Introduction

In this report, the diagnostic extension of the FESTO Processing Station is presented. The station makes a basic industrial automation routine where a part is located in the socket, the test socket/table transports it, the part is drilled, checked by the testing cylinder, and taken to its final destination.

The program of automatic operation of the station already worked successfully. The task was to introduce a diagnostic system for possible errors and notify the operator about them via indicator lamps.

The diagnostic extension was designed for the older FESTO/FST software that ran on Windows XP. For this reason, the program was created using simple PLC logic, timers, flags, and comparison blocks without any special libraries.

2. Executive Summary

Five types of possible errors were programmed for the FESTO Processing Station. They correspond to the most likely failures of the equipment: no material in the socket, test socket transport error, drill cylinder error, testing cylinder error, and stop/ emergency stop condition.

Moreover, an additional function of diagnostics override was introduced. Once the error occurred during the cycle after its automatic start, the operator can press the button Input I2.2 to bypass the current diagnostic and go to the next step. Override is impossible if the automatic cycle is yet to start, and it also cannot be done for stop/e-stop conditions.

No.	Diagnostic Category	Root Cause Domain	Detection Type	Lamp Indication
1	No Part Detected at Start	Process / Material	Instantaneous	O1.0 blinking
2	Test Socket Position Fault	Mechanical / Electrical / Sensor	Timer-based	O1.0 + O1.1 blinking
3	Drill Cylinder Fault	Pneumatic / Mechanical / Sensor	Timer-based + logical conflict	O1.0 + O1.2 blinking
4	Testing Cylinder Fault	Pneumatic / Mechanical / Sensor	Timer-based + logical conflict	O1.1 + O1.2 blinking
5	Stop or Emergency Stop Condition	Operator / Safety	Instantaneous	O1.1 blinking

3. Station Description

The major components of the FESTO Processing Station include: socket, drilling slot, drill, test socket and testing cylinder, and final slot.

Label	Station Part	Function
A	Part taking slot / socket	Initial position where the part is detected
B	Drilling slot	Position where the part is held and drilled
C	Drill	Performs the drilling operation
D	Test socket / testing cylinder	Checks the drilled hole
E	Final slot	Final output position of the processed part

A basic cycle starts when a part is put in the socket and the operator presses the Start button. Then, the test socket/table starts transporting it to the drilling slot, and when it is finished, the drill starts working. At first, the drill cylinder goes down, then the drilling process is done, and finally, the drill cylinder returns back to its upper position. The table transports the part to the testing slot, and the testing cylinder checks it. Then, the table transports the part to the final slot.

4. Allocation List

4.1. Inputs

Address	Description
I0.0	Part in socket
I0.1	Test socket turn position sensor
I0.2	Hold cylinder in home position
I0.3	Hold cylinder in final position
I0.4	Drill cylinder in upper position
I0.5	Drill cylinder in lower position
I0.6	Testing cylinder in upper position
I0.7	Testing cylinder in lower position
I2.0	Start button
I2.1	Reset / clear diagnostics
I2.2	Positioning button / diagnostic override
I2.3	Auto / manual control
I2.4	Stop button
I2.7	Reset emergency stop

4.2. Outputs

Address	Description
O0.1	Drill engine
O0.2	Test socket engine
O0.3	Drill position down
O0.4	Drill position up
O0.5	Hold piston to home position
O0.6	Testing piston to lower position
O0.7	Piston holder
O1.0	Start lamp / diagnostic LED 1
O1.1	Reset lamp / diagnostic LED 2
O1.2	Positioning lamp / diagnostic LED 3

4.3. Added Diagnostic Flags

Flag	Description
F0.0	Diagnostic 1: No part detected
F0.1	Diagnostic 2: Test socket position fault

F0.2	Diagnostic 3: Drill cylinder fault
F0.3	Diagnostic 4: Testing cylinder fault
F0.4	Stop flag
F0.5	Diagnostic 5: Stop or emergency stop condition
F0.6	Diagnostic blink OFF phase
F0.7	Diagnostic blink ON phase

4.4. Added Override Flags

Flag	Description
F1.0	Automatic cycle started / override allowed
F1.1	Override return to Step 8
F1.2	Override return to Step 16
F1.3	Override return to Step 18
F1.4	Override return to Step 25
F1.5	Override return to Step 31
F1.6	Override return to Step 1

4.5. Added Timers

Timer	Description
T7	Diagnostic LED ON timer
T8	Diagnostic LED OFF timer
T9	Diagnostic movement fault timeout

5. Reset and Diagnostic Recovery Procedure

Pressing the Reset button I2.1 turns off the diagnostic lamps, removes active diagnostic flags and blinking lamps, resets the timers, and resets the return override flags F1.1–F1.6.

It allows the operator to clear up the diagnostic code.

```
IF I2.1 'Reset / clear diagnostics
THEN RESET F0.0 'DIAG 1 - no part
     RESET F0.1 'DIAG 2 - test socket
     RESET F0.2 'DIAG 3 - drill cylinder
     RESET F0.3 'DIAG 4 - testing cylinder
     RESET F0.5 'DIAG 5 - stop or emergency
     RESET F0.6 'Diagnostic blink OFF phase
     RESET F0.7 'Diagnostic blink ON phase
     RESET F1.0 'Automatic cycle started / override allowed
     RESET F1.1 'Override return to step 8
     RESET F1.2 'Override return to step 16
     RESET F1.3 'Override return to step 18
     RESET F1.4 'Override return to step 25
     RESET F1.5 'Override return to step 31
     RESET F1.6 'Override return to step 1
     RESET O1.0 'Start LED
     RESET O1.1 'Reset LED
     RESET O1.2 'Positioning LED
     RESET T7
     RESET T8
     RESET T9
```

6. Diagnostic Override Function

Additional diagnostic function was implemented using the input I2.2.

Normally, it serves to start positioning program. But once the automatic cycle is launched, it may work as a diagnostics override button.

It works depending on the state of the diagnostic flag F1.0. If it is not active (off), the automatic cycle has not started, and the override is not possible. If F1.0 is active, the override is permitted.

Thus, it prevents the operator from continuing processing without a part at the beginning, and also from overriding the stop and e-stop conditions.

Condition	I2.2 Function
Automatic cycle has not started	Normal positioning
Automatic cycle has started and a timeout fault occurs	Ignore fault and continue to stored next step
Emergency stop or Stop condition active	Override not allowed
No part detected before cycle starts	Override not allowed

Flag	Meaning
F1.0 = 0	Automatic cycle has not started, override is not allowed
F1.0 = 1	Automatic cycle has started, override is allowed for selected faults

7. Detailed Fault Analysis

7.1. Fault 1: No Part Detected at Start

Root Cause Domain: Process / Material

Severity: High

Detection Type: Instantaneous

Diagnostic Flag: F0.0

Lamp Indication: O1.0 blinking

The fault happens when the operator presses the start button, but the sensor I0.0 fails to detect a part in the socket.

Since there is no part, the automatic operation cannot safely proceed because it involves rotating, drilling and testing.

Root causes might include: no part in the socket, wrong part positioning, malfunctioning or dirty sensor I0.0, loose/damaged wires of the sensor, part too small to be detected by the sensor or faulty sensor input.

In order to solve the problem, the operator should ensure that there is a part in the socket and that it is well positioned. Then the operator should check the LED of I0.0 and clean the sensor if needed. Also, the wiring and connector should be checked. Then press reset I2.1 and Start I2.0 once the part is detected.

7.1.1. Detection Mechanism

Parameter	Details
Input checked	I0.0 Part in socket
Trigger	I2.0 Start button pressed
Fault condition	I2.0 ON and I0.0 OFF
Diagnostic flag	F0.0
Indicator	O1.0 blinking
Override allowed	No

7.1.2. Code Implementation

```
IF I2.0 'Start
   AND I2.7 'Emergency is reset
```

```
   AND N I0.0 'Part in socket
THEN SET F0.0 'DIAG 1 - no part detected
     RESET F0.1
     RESET F0.2
     RESET F0.3
     RESET F0.5


IF I2.0 'Start
   AND I2.7 'Emergency is reset
   AND I0.0 'Part in socket
THEN SET F1.0 'Automatic cycle started / override allowed
     SET P2 'AUTOMATIC
```

7.2. Fault 2: Test Socket Position Fault

Root Cause Domain: Mechanical / Electrical / Sensor

Severity: High

Detection Type: Timer-based

Diagnostic Flag: F0.1

Lamp Indication: O1.0 + O1.1 blinking

The fault occurs when engine O0.2 gets energized, but sensor I0.1 fails to become active within the set time.

The table/test socket must reach its right position for further drilling/testing operation to proceed. Thus, if sensor I0.1 fails to confirm the right position, the process becomes out-of-order.

Root causes for the problem might include: jamming of the test socket/table, malfunctions in the motor or actuator which is operated with O0.2, malfunction in the position sensor I0.1, malfunction in the wiring and connector of I0.1, incomplete rotation of the test socket/table, insufficient power supplied to O0.2 and sensor I0.1 or mechanical blockage in the way of the table/test socket.

To solve the problem, the operator should check the physical rotation of the test socket/table, check whether the path is blocked, verify O0.2 with its indicator light and verify I0.1 with the help of its LED. The operator should also check whether I0.1 activates when test/socket comes into its position. The cable connecting the sensor and the PLC should also be checked. Once the problem is diagnosed, press Reset I2.1. Then press Override I2.2 to allow continuing if the process is already in motion.

7.2.1. Detection Mechanism

Parameter	Details

Output command	O0.2 Test socket engine
Sensor feedback	I0.1 Test socket turn position sensor
Timer used	T9
Timeout value	5 seconds
Fault condition	T9 finished and I0.1 still OFF
Diagnostic flag	F0.1
Indicator	O1.0 + O1.1 blinking
Override allowed	Yes, only if automatic cycle already started

7.2.2. Code Implementation

```
IF N T1 'FIRST DELAY
THEN SET O0.2 'Test socket engine
     SET T9 'Diagnostic movement fault timeout
       WITH 5s


IF N T9 'Diagnostic movement fault timeout
   AND N I0.1 'Test socket turn position sensor
THEN RESET F0.0
     SET F0.1 'DIAG 2 - test socket fault
     RESET F0.2
     RESET F0.3
     RESET F0.5
     RESET O0.2 'Test socket engine
     SET F1.1 'Override return to step 8
     JMP TO 200
```

7.3. Fault 3: Drill Cylinder Fault

Root Cause Domain: Pneumatic / Mechanical / Sensor

Severity: Critical

Detection Type: Timer-based and logical sensor conflict

Diagnostic Flag: F0.2

Lamp Indication: O1.0 + O1.2 blinking

The fault occurs when the cylinder fails to come to its position within the set time. The other possible fault is simultaneous activation of the upper and lower sensor.

It should never happen that the upper and the lower position sensors get activated simultaneously since it means some wiring or sensor problems.

The root cause for the fault might include: jamming of the cylinder, low pressure of the air in the drill chamber, malfunction of the solenoid valve O0.3/O0.4, malfunction of I0.4 or I0.5 position sensors, damage to wires from these sensors, slow moving cylinder, mechanical blocking in the drill cylinder or electrical short circuit resulting in activation of both position sensors.

Solving the problem involves: checking the air pressure supply in the drill cylinder, checking the mechanical condition of the drill cylinder. Also, the output O0.3/O0.4 should be verified. Moreover, the LEDs on I0.4 and I0.5 should be verified and ensure that only one position sensor works at any time. The cable should be inspected and any mechanical blockage be removed. After diagnostics, press Reset I2.1. If the automatic cycle had been initiated and the problem was timeout only, I2.2 can be used for override.

7.3.1. Detection Mechanism

Parameter	Details
Down command	O0.3 Drill position down
Up command	O0.4 Drill position up
Lower position sensor	I0.5
Upper position sensor	I0.4
Timer used	T9
Timeout value	3 seconds
Fault condition 1	Drill down commanded but I0.5 not reached
Fault condition 2	Drill up commanded but I0.4 not reached
Fault condition 3	I0.4 and I0.5 active simultaneously
Diagnostic flag	F0.2
Indicator	O1.0 + O1.2 blinking
Override allowed	Yes, only if automatic cycle already started and not a safety stop

7.3.2. Code Implementation

```
IF O0.1 'Drill engine
THEN SET O0.3 'Drill position down
     SET T3 'DRILLING TIMER
       WITH 2s
     SET T4 'DRILL TIMER BUFFER
       WITH 1s
     SET T9 'Diagnostic movement fault timeout
       WITH 3s

IF N T9 'Diagnostic movement fault timeout
```

```
   AND N I0.5 'Drill cylinder in lower position
THEN RESET F0.0
     RESET F0.1
     SET F0.2 'DIAG 3 - drill cylinder lower fault
     RESET F0.3
     RESET F0.5
     RESET O0.1 'Drill engine
     RESET O0.3 'Drill position down
     SET F1.2 'Override return to step 16
     JMP TO 200


IF N T9 'Diagnostic movement fault timeout
   AND N I0.4 'Drill cylinder in upper position
THEN RESET F0.0
     RESET F0.1
     SET F0.2 'DIAG 3 - drill cylinder upper fault
     RESET F0.3
     RESET F0.5
     RESET O0.1 'Drill engine
     RESET O0.4 'Drill position up
     SET F1.3 'Override return to step 18
     JMP TO 200


IF I0.4 'Drill cylinder in upper position
   AND I0.5 'Drill cylinder in lower position
THEN RESET F0.0
     RESET F0.1
     SET F0.2 'DIAG 3 - drill sensor contradiction
     RESET F0.3
```

7.4. Fault 4: Testing Cylinder Fault

Root Cause Domain: Pneumatic / Mechanical / Sensor

Severity: High

Detection Type: Timer-based and logical sensor conflict

Diagnostic Flag: F0.3

Lamp Indication: O1.1 + O1.2 blinking

This fault happens when the testing cylinder is requested to move downwards, but the lower position sensor I0.7 is not detected within the acceptable period.

Testing cylinder is used for checking the drilled hole or confirmation of processed part. If the cylinder cannot reach the required position, there will be no confirmation of inspection performed.

The two testing cylinders sensors must never be active at the same time.

Possible faults include jammed testing cylinder, insufficient pneumatic pressure, broken or disconnected solenoid, clogged testing probe due to wrong positioning of a part, malfunctioning lower position sensor I0.7, wiring fault on one of sensors (I0.6, I0.7), both sensors detecting due to short circuit or misalignment.

To fix this problem, the operator needs to inspect the movement of testing cylinder and inspect the testing probe for any obstructions. The lower position sensor I0.7 and upper position sensor I0.6 should be inspected. It is also necessary to make sure that both sensors are never active simultaneously. Pneumatic tubing and valve need to be inspected as well. Once the problem is fixed, Reset I2.1 button needs to be pushed. The override signal I2.2 can be used if it is safe to continue the automatic cycle that has already started.

7.4.1. Detection Mechanism

Parameter	Details
Output command	O0.6 Testing piston to lower position
Upper position sensor	I0.6
Lower position sensor	I0.7
Timer used	T9
Timeout value	3 seconds
Fault condition 1	O0.6 commanded but I0.7 not reached
Fault condition 2	I0.6 and I0.7 active simultaneously
Diagnostic flag	F0.3
Indicator	O1.1 + O1.2 blinking
Override allowed	Yes, only if automatic cycle already started and not a safety stop

7.4.2. Code Implementation

```
IF N T2 'MOVEMENT DELAY
THEN SET O0.6 'Testing piston to lower position
     SET T4 'DRILL TIMER BUFFER
       WITH 1s
     SET T9 'Diagnostic movement fault timeout
       WITH 3s

IF N T9 'Diagnostic movement fault timeout
```

```
   AND N I0.7 'Testing cylinder in lower position
THEN RESET F0.0
     RESET F0.1
     RESET F0.2
     SET F0.3 'DIAG 4 - testing cylinder lower fault
     RESET F0.5
     RESET O0.6 'Testing piston to lower position
     SET F1.5 'Override return to step 31
     JMP TO 200

IF I0.6 'Testing cylinder in upper position
   AND I0.7 'Testing cylinder in lower position
THEN RESET F0.0
     RESET F0.1
     RESET F0.2
     SET F0.3 'DIAG 4 - testing sensor contradiction
```

7.5. Fault 5: Stop or Emergency Stop Condition

Root Cause Domain: Operator / Safety

Severity: Critical

Detection Type: Instantaneous

Diagnostic Flag: F0.5

Lamp Indication: O1.1 blinking

This fault is triggered either by pressing of Stop button or lack of emergency stop reset condition.

This situation is considered a safety condition. The primary reason of displaying this warning is not merely diagnostics but making sure that operator inspects the machine before automatic cycle continues.

Root causes of the problem include pressing Stop during cycle, activation of emergency stop, absence of emergency stop reset condition, interruption of operation by operator due to unsafe condition, or machine condition requiring operator to perform some safety checks before operation can resume.

To resolve the fault, it is necessary to identify the reasons for pressing Stop and/or emergency stop condition activation. Any obstructions and other safety concerns should be resolved. Motion of the machine must be stopped completely by the operator before anything else. The operator needs to ensure that emergency stop circuit is released. After this, Reset I2.1 button needs to be pushed. Overriding of this diagnosis is explicitly forbidden.

7.5.1. Detection Mechanism

Parameter	Details
Stop input	I2.4
Emergency reset input	I2.7
Diagnostic flag	F0.5
Indicator	O1.1 blinking
Override allowed	No

7.5.2. Code Implementation

```
IF I2.4 'Stop
THEN SET F0.4 'FLAG
     SET F0.5 'DIAG 5 - stop pressed / start required
     RESET F1.0 'Automatic cycle started / override allowed


IF N I2.7 'Reset emergency stop
THEN RESET F0.0
     RESET F0.1
     RESET F0.2
     RESET F0.3
     SET F0.5 'DIAG 5 - emergency stop / reset required
     RESET O0.1
     RESET O0.2
     RESET O0.3
     RESET O0.4
     RESET O0.5
     RESET O0.6
     RESET O0.7
     RESET F1.0
```

8. Diagnostic LED Control

LED diagnostics is implemented in the compare block CMP2. There are three LEDs available which are blinking with the period of 1 second on and 1 second off.

The basic principle is that the diagnostic flag is raised, then the beginning of ON phase begins with F0.7 signal. Necessary combination of LEDs goes ON, then the ON phase ends after time period T7 and the LEDs go off. Afterward, F0.6 initiates the OFF phase which is terminated by T8 and cycle starts again.

This makes the problem more obvious since there is blinking of lights rather than constant lighting.

Timer / Flag	Function
T7	LED ON phase timer
T8	LED OFF phase timer
F0.7	Blink ON phase
F0.6	Blink OFF phase

9. Lamp Indication Summary

Fault No.	Fault Title	Active Flag	Lamp Pattern	First Inspection Area
1	No part detected at start	F0.0	O1.0 blinking	Part socket, I0.0 sensor
2	Test socket position fault	F0.1	O1.0 + O1.1 blinking	Table/socket movement, I0.1 sensor
3	Drill cylinder fault	F0.2	O1.0 + O1.2 blinking	Drill cylinder, I0.4/I0.5 sensors
4	Testing cylinder fault	F0.3	O1.1 + O1.2 blinking	Testing cylinder, I0.6/I0.7 sensors
5	Stop or emergency stop	F0.5	O1.1 blinking	Stop button, emergency circuit

10. Diagnostic Stop State

If diagnostic problem appears during the automatic cycle, the program goes to Step 200. This ensures the disabling of all harmful and moving outputs so that the automatic cycle will not progress further without solving the problem.

Thus, the diagnostic state becomes much safer since there will be turning off of motion commands before any overriding attempts.

```
STEP 200
IF NOP
THEN RESET O0.1 'Drill engine
     RESET O0.2 'Test socket engine
     RESET O0.3 'Drill position down
     RESET O0.4 'Drill position up
     RESET O0.6 'Testing piston to lower position
     RESET T9 'Diagnostic movement fault timeout
     JMP TO 201
```

11. Diagnostic Override / Ignore Logic

Diagnostic override logic is introduced at Step 201.

In case of any fault, an operator may choose to press Reset I2.1 to clear diagnostics and restart the program from Step 1 or press Positioning I2.2 to ignore the diagnostic fault and continue the program to the stored next step.

Nevertheless, Positioning I2.2 will serve as override only when F1.0 flag is active, which implies that the automatic cycle has been launched.

Specifically, in case of test socket fault in the earlier stage than Step 8, the F1.1 is activated by the program. Upon pressing the Positioning I2.2 button, the diagnostic fault is cleared and program moves on to Step 8.

The above scheme is repeated for other stored return points as well.

Override Flag	Return Step
F1.1	Step 8
F1.2	Step 16
F1.3	Step 18
F1.4	Step 25
F1.5	Step 31
F1.6	Step 1

11.0.1. Override Example Code

```
IF I2.2 'Positioning button used as diagnostic override
   AND F1.0 'Automatic cycle started / override allowed
   AND F1.1 'Stored return step
   AND N F0.5 'Do not override stop/emergency
THEN RESET F0.0
     RESET F0.1
     RESET F0.2
     RESET F0.3
     RESET F0.6
     RESET F0.7
     RESET O1.0
     RESET O1.1
     RESET O1.2
     RESET T7
     RESET T8
     RESET T9
```

```
RESET F1.1
RESET F1.2
RESET F1.3
RESET F1.4
RESET F1.5
RESET F1.6
JMP TO 8
```

12. Program Files Modified

File	Purpose of Change
P0_MAIN_DIAGNOSTICS_OVERRIDE	Added reset logic, start condition check, no-part diagnostic, and positioning/override separation
P2_AUTOMATIC_DIAGNOSTICS_OVERRIDE	Added movement timeout diagnostics, diagnostic stop state, and override return logic
CMP0_emergency_DIAGNOSTICS_OVERRIDE	Added emergency diagnostic and clears override flags
CMP1_STOP_DIAGNOSTICS_OVERRIDE	Added stop diagnostic and clears override flags
CMP2_DIAGNOSTICS	Added LED blinking logic and sensor contradiction checks
ALOCATION LIST_DIAGNOSTICS_OVERRIDE	Added diagnostic flags, timers, and override flag descriptions

13. Testing and Validation Procedure

Validation of diagnostic functions took place by creating each of the mentioned faults and observing the behavior of corresponding flag, lamp pattern, and program flow. Test cases included station operation without the part, obstruction or disconnection of position sensors, inability of cylinders to reach the specified position, creation of contradictive state of the sensors, activation of Stop or Emergency Stop faults, and testing override in case of fault due to timeout.

Correct response was assumed when the proper diagnostic flag was activated, corresponding lamp pattern began blinking, and potentially dangerous outputs were deactivated if required.

Override logic was verified by making sure that only faults after starting of the automatic cycle could be ignored and that no override was possible before cycle start or during stop/e-stop situations.

Test No.	Test Scenario	Expected Result
1	Press Start with no part in socket	F0.0 activates, O1.0 blinks, automatic cycle does not start
2	Block or disconnect I0.1 during table rotation	F0.1 activates, O1.0 + O1.1 blink, movement stops
3	Prevent drill lower sensor I0.5 from activating	F0.2 activates, O1.0 + O1.2 blink, drill command stops
4	Prevent drill upper sensor I0.4 from activating	F0.2 activates, O1.0 + O1.2 blink, drill up command stops
5	Prevent testing lower sensor I0.7 from activating	F0.3 activates, O1.1 + O1.2 blink, testing command stops
6	Force I0.4 and I0.5 ON together	F0.2 activates as drill sensor contradiction
7	Force I0.6 and I0.7 ON together	F0.3 activates as testing sensor contradiction
8	Press Stop during automatic cycle	F0.5 activates, O1.1 blinks, override is not allowed
9	Activate emergency stop condition	F0.5 activates, O1.1 blinks, outputs are reset
10	Press I2.2 after a timeout fault during cycle	Fault clears and program continues to stored next step
11	Press I2.2 before automatic cycle starts	Positioning function works normally; diagnostic override is not active

14. Safety Notes

Diagnostic override can be useful for demonstration purposes or in case of intentional creation of faults during lab testing and for proceeding through the process once the malfunctioning sensor is identified.

It is critical to remember that if a cylinder does not reach its specified position, overriding will lead to a collision or incorrect drilling and testing operations.

Thus, no-part-at-start fault cannot be overridden, stop and e-stop faults cannot be overridden, reset is the only appropriate reaction in most cases, and override shall be performed only when operator knows about the safety of the next step.

15. Conclusion

In conclusion, this report described the implementation of the diagnostic system for the FESTO Processing Station. In addition to the existing automatic sequence, five additional diagnostic categories related to material absence, table positioning problem, drill cylinder fault, testing cylinder fault, and stop/e-stop conditions were added to the system.

Diagnosis was implemented utilizing all existing PLC resources, namely input sensors, output lamps, flags, timers, and compare blocks. Three existing output lamps were used for diagnostic purposes by changing their blinking patterns (O1.0, O1.1, and O1.2).

A controlled diagnostic override using Positioning button (I2.2) was also implemented to allow selected faults to be ignored only after the launch of the automatic cycle.

Overall, diagnostics greatly improve fault detection capability and provide valuable information for troubleshooting and system safety, while being fully compatible with older FESTO/FST4 environment and Windows XP-based programming.